A certified three-way triage gate for industrial visual inspection: escaped-defect and false-reject guarantees on MVTec AD, VisA, and MPDD

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Abstract

Anomaly detectors reach near-ceiling image-AUROC on MVTec AD, yet AUROC is not a deployable decision: it names no operating threshold and bounds neither asymmetric-cost error — passing a defective part (an *escaped defect*) or rejecting a good one (a *false reject*). We present *inspect-gate*, a distribution-free triage layer sending each image to auto-pass, auto-reject, or defer on any backbone's scores, with per-category certified escaped-defect ($P(\text{auto-pass} \mid \text{defective}) \leq \alpha_{\text{miss}}$) and false-reject ($P(\text{auto-reject} \mid \text{good}) \leq \alpha_{\text{fr}}$) rates. Both reduce exactly to split-conformal calibration — false-reject directly, escaped-defect through a score-negation order-statistic identity — adding no new estimator, inheriting finite-sample validity. A category whose calibration pool is too small is *refused* and reported audited-not-certified, never rounded. Across two backbones and five seeds, the machinery behaves as predicted, no kill-gate tripped. On MVTec AD, escaped-defect certifies in all 15 categories and false-reject in the four with enough good-calibration data; a preregistered train-holdout remedy (PatchCore only) lifts false-reject to 12–13 of 15, KS-failing categories audited-not-certified. On VisA both axes certify in all 12 categories, the validity audit passes non-vacuously in all 120 cells, and the deferral-skill audit becomes informative. A third benchmark, MPDD (painted-metal parts), is the stingy extreme: its uniformly small good pools certify false-reject in 0/6 categories at the primary protocol (escaped-defect still 5/6), and a flag-gated train-holdout arm rescues 4/6, with one KS-audited out, one below the calibration-pool floor — refusal tracking the data throughout (0/6 $\rightarrow$ 4/15 $\rightarrow$ 12/12). The confirmatory pooled-category

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audit returns a constructive verdict on MVTec AD.

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Highlights

- A distribution-free gate routes inspection images to pass, reject, or defer.
- Per-category certified escaped-defect and false-reject rates from conformal.
- Refusal is a first-class output when a good-calibration pool is too small.
- Validated on MVTec AD, VisA, and MPDD, two backbones and five seeds each.
- Adds 2.8 microseconds per image; certification is free relative to detection.

1. Introduction

Automated visual inspection is one of the most mature industrial applications of computer vision, and unsupervised anomaly detection (AD) on the MVTec AD benchmark [1] is its standard proving ground. Modern detectors — memory-bank methods such as PatchCore [2] and reconstruction/transformer methods such as Dinomaly [3] — report image-level AUROC above 0.98 and, for the newest models, above 0.99. On the usual reading the problem looks solved.

It is not solved as a decision. Image-AUROC is a threshold-free ranking summary: it says a detector orders defective above good images well, but it names no operating point, and on a real line an operating point is exactly what is needed. Worse, the two mistakes a threshold can make are not symmetric. Passing a defective part (an *escaped defect*) can ship a fault to a customer; rejecting a good part (a *false reject*) wastes yield and, at high rates, risks eroding operator trust in the system. A useful inspection system must bound both, per product category, with a guarantee that holds in finite samples rather than on average over an idealized distribution.

We take the position that the deployable object is not a better detector but a certified *triage layer* on top of an existing one. Concretely, we route each image into three actions: *auto-pass* (confident-good, ship without review), *auto-reject* (confident-defective, scrap or rework without review), and *defer* (send to a human). The value proposition is that the two automated actions carry certificates: the auto-pass region is calibrated so that the probability a truly defective image lands there is at most α_{miss}, and the auto-reject region so that the probability a truly good image lands there is at most α_{fr}; everything else defers. The human-review budget is then whatever the data forces, and — crucially — it is visible: a category whose data cannot support the requested guarantee defers more, rather than pretending to a certificate it cannot honor. We evaluate the layer on three complementary benchmarks by design: MVTec AD, whose small per-category good pools stress the refusal machinery; VisA, whose lower score ceilings and larger good pools let both certificates bind everywhere and give the deferral-skill audit real headroom; and MPDD (real painted-metal-parts fabrication), whose uniformly small good pools make it the stingiest certifiability point — false-reject certifiable in 0/6 categories under the primary protocol, the extreme that shows the gate refuses precisely when the data cannot pay for a certificate.

This framing turns a benchmark score into an auditable decision and makes three contributions.

- **A certified three-way triage system for inspection.** The contribution is not a new estimator — both guarantees reduce exactly to split-conformal quantiles (Section 3.1: false-reject directly, escaped-defect through a score-negation order-statistic identity) — but the coupled system assembled from them: two simultaneously certified error axes (escaped-defect and false-reject), explicit per-category certifiability floors, first-class refusal, and the K1–K7 kill-gate battery, packaged as an auditable decision layer over any backbone's scores. To our knowledge no published method delivers this exact combination on MVTec AD or VisA; the binding K6 pre-submission citation re-scan, executed 2026-07-13 (Section 2), returned a clear verdict — no scoop. We foreground the "no new math" property deliberately: it is what makes the guarantees finite-sample-exact and the tool a thin, testable wrapper rather than a heuristic.

- **Refusal as a first-class outcome, with teeth.** A per-category certifiability floor $\alpha_{\min} = 1/(n_{\text{cal}} + 1)$ makes "we cannot certify this category at this rate" an explicit, reported state (audited-not-certified), never a silently loosened threshold. This is not cosmetic: on our own substrate (Table 2), a non-refusing baseline that issues an auto-reject threshold and claims the requested false-reject rate $\alpha_{\text{fr}} = 0.05$ in the 11 categories the gate refuses would be asserting a guarantee the calibration pool cannot support — the achievable floor there is 0.06–0.14 (e.g. toothbrush $1/7 = 0.14$), 1.2×–2.9× looser than claimed — so it silently over-promises where the gate refuses and defers. (On the escaped-defect axis the floor clears $\alpha_{\text{miss}} = 0.10$ in all 15 categories, so this failure mode does not arise there; a post-hoc binding demonstration (Section 5.9) confirms this empirically — in a range of (backbone, category) cells a non-refusing fixed-threshold baseline's *realized* escaped-defect or false-reject rate exceeds the target while the certified gate holds within it on the same data.)

- **A practitioner-facing audit.** Beyond the certificate, we test whether field-standard threshold practice earns any deferral skill over honest random deferral (an excess-AURC audit), reporting a constructive or a debunking verdict symmetrically; the preregistered confirmatory pooled audit returns the constructive verdict — practice earns real deferral skill in all five seeds (Section 5.8).

Figure 1 summarizes the resulting pipeline, from a frozen backbone's per-image score through the G1/G2 certificates and certifiability floor to the three-way auto-pass/defer/ auto-reject route.

We are explicit about status. The preregistration was frozen on 2026-07-11 (sha256-pinned sign-off; Section 6); this manuscript reports the primary-protocol axes on all three benchmarks plus the frozen G2 train-holdout promotion arm and two *post-freeze* benchmarks (VisA and MPDD — neither part of the frozen preregistration; both run under the identical frozen protocol and code path, their audits reported as exploratory and never in the confirmatory Holm family), and marks every remaining preregistered-but-uncomputed arm as an explicit slot. No uncomputed result is presumed anywhere in the text.

2. Related work

Distribution-free risk control.. The gate is an application of conformal prediction [4, 5, 6]: given an exchangeable calibration set, an order statistic of the calibration scores yields a threshold with a finite-sample coverage guarantee. The specific ingredient we use is the split (inductive) conformal construction [7], whose single calibration/estimation split makes it cheap enough to apply per category at deployment time. Set-valued and adaptive variants of the same idea — least-ambiguous set-valued classifiers with a bounded error level [8] and adaptive-coverage prediction sets [9] — show how

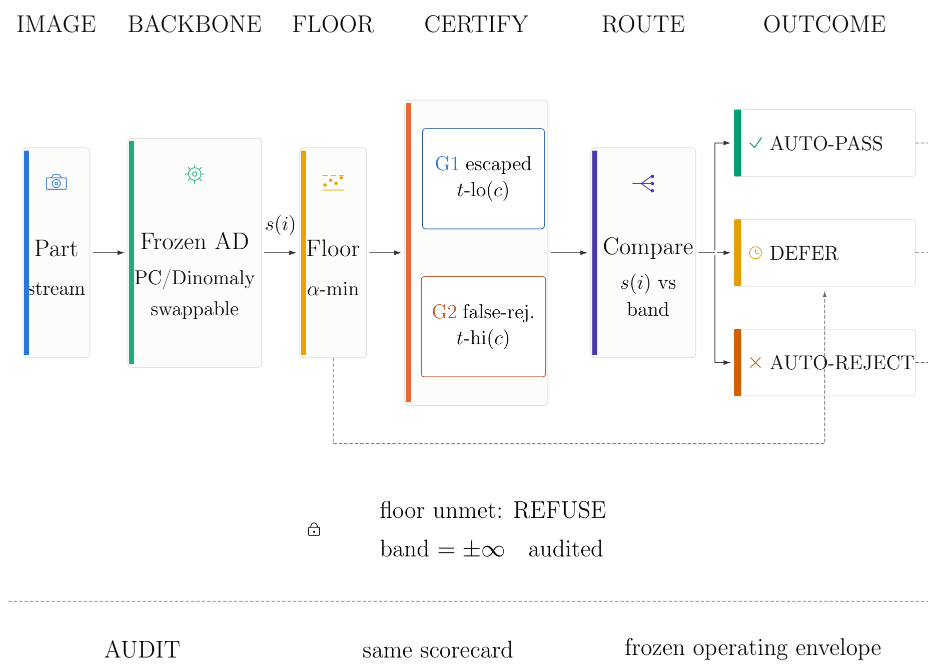

Figure 1: Method overview. A frozen backbone (PatchCore or Dinomaly, interchangeable) scores each image; per-category calibration checks the certifiability floor $\alpha_{\min} = 1/(n_{\mathrm{cal}}+1)$ (Section 3.2) and, when met, sets the G1 escaped-defect threshold $t_{\mathrm{lo}}(c)$ and the G2 false-reject threshold $t_{\mathrm{hi}}(c)$ (Section 3.1). The three-way route then compares each score against $(t_{\mathrm{lo}}, t_{\mathrm{hi}})$ to auto-pass, defer, or auto-reject, carrying the stated escaped-defect and false-reject guarantees; an unmet floor instead refuses the axis (band $= \pm\infty$), defers, and is reported audited-not-certified.

a one-sided error target maps onto a calibrated threshold, which is exactly the primitive our two certificates specialise. The broader risk-controlling line — risk-controlling prediction sets [10] and Learn-then-Test [11] — generalizes this to bounding arbitrary monotone risks and to multiple-hypothesis calibration. All of these guarantees are conditional on exchangeability between calibration and deployment data; conformal prediction under covariate shift [12] characterises what breaks when that assumption fails, and is the lens through which we frame threshold-transfer across production periods as future work (Section 6). Our escaped-defect and false-reject bounds are the two one-sided special cases that split conformal already delivers exactly (Section 3.1); we deliberately use the simplest sufficient construction rather than the more general RCPS/LTT machinery, and cite the latter as the family the method belongs to.

Industrial anomaly detection.. MVTec AD [1] defined the per-category unsupervised setting we use, and the field it seeded is now broad enough to warrant dedicated surveys [13]. The dominant families all emit an image- or pixel-level anomaly score our gate can consume unchanged: memory-bank and feature-distribution methods such as PatchCore [2], PaDiM [14], and SPADE [15]; student–teacher and knowledge-distillation methods such as uninformed students [16] and reverse distillation [17]; reconstruction and synthetic-defect methods such as DRÆM [18] and CutPaste [19]; normalizing-flow methods such as FastFlow [20]; and recent simplified or transformer models such as SimpleNet [21] and Dinomaly [3]. Of these, PatchCore and Dinomaly are the two reproduced backbones here; EfficientAD [22] is a fast alternative we discuss but do not certify (its only public implementations are community reimplementations, so it cannot anchor a binding reproduction gate). Dinomaly descends from the multi-class (one model, many categories) line initiated by UniAD [23] — the setting its published VisA figure reports and the one we use on VisA (Section 4). Within the engineering-informatics literature specifically, this detector line has been carried directly into deployable inspection systems: component-aware anomaly detection for logical industrial inspection [24], defect-aware transformer networks for visual surface-defect detection [25], and autoencoder-based surface anomaly inspection [26], alongside segmentation-based deep-defect detection [27] and broader automated-visual-inspection surveys [28]. That body of work optimises the *detector*; our contribution is the orthogonal decision layer that sits on top of any such detector's scores and attaches a certified routing guarantee. MPDD [29] adds a real painted-metal-parts fabrication benchmark whose non-homogeneous backgrounds and small good pools make it a deliberately harsher industrial-realism stress test than the two academic staples — though, like them, still an academic benchmark rather than production-line data (see Limitations, Section 6). We use anomalib [30] for the PatchCore implementation. Our contribution is orthogonal to all of these: we consume their scores and add a decision layer, and we improve none of them.

Selective prediction and abstention.. The defer action is a selective-prediction mechanism [31, 32], but applied at the decision layer with a two-sided, per-category certificate rather than a single accuracy-coverage curve or a learned reject head; the auto-reject certificate (false-reject control) has no analogue in standard selective classification, which abstains only around the positive decision.

Conformal anomaly detection and abstention.. Conformal methods have been applied to anomaly detection before, but not to the certified inspection-triage decision we target. Early work bounded false-alarm rates for sequential and trajectory anomalies [33, 34]; recent tooling controls the false-discovery rate over flagged anomalies [35]; conformal risk control [36] and its graph instantiation CRC-SGAD [37] bound a monotone risk (e.g. FNR/FPR) but emit a single flag, not a three-way route; and dual-threshold conformal abstention for autonomous-system perception [38] adds a reject option on camera/LiDAR (and classification) tasks, with no industrial-inspection estimand and no evaluation on MVTec AD. The closest defect-detection application [39] bounds a single mean error rate for surface-defect segmentation, again with no deferral option and not on MVTec AD. To our knowledge no published method delivers a certified *three-way* triage (auto-pass / auto-reject / defer) on MVTec AD with *coupled* escaped-defect and false-reject guarantees and explicit per-category certifiability floors; the preregistered K6 scoop gate, run as a fresh pre-submission re-scan on 2026-07-13, confirms this — the closest published systems [38, 37, 39] each differ on a load-bearing axis (substrate, estimand coupling, or the defer option), and two adjacent 2026 works (a selective-risk certificate on generic vision, and a conformal cyber-physical-systems detector on sensor time-series) target neither the image-inspection estimand nor the three-way route. That is the seam this paper occupies. Section 5.11 instantiates the single-threshold CRC construction [36] as a quantitative baseline on our own canonical score dumps under the identical protocol, isolating what the three-way route and its coupled false-reject certificate add over the nearest single-flag alternative. More broadly, a recurring reliability finding is that low aggregate error coexists with brittle or miscalibrated decisions; our escaped-defect vs. false-reject framing is the industrial-inspection instance of that gap, and the excess-AURC audit (Section 3.4) tests whether standard thresholding beats chance at spending a deferral budget.

3. Methods

3.1. The three-way gate and its two certificates

Each image i in category c has an anomaly score s_i with the convention *higher = more anomalous*. The gate is a pair of per-category thresholds $(t_{\mathrm{lo}}(c), t_{\mathrm{hi}}(c))$ with $t_{\mathrm{lo}} \leq t_{\mathrm{hi}}$: images with $s_i \leq t_{\mathrm{lo}}(c)$ auto-pass, images with $s_i \geq t_{\mathrm{hi}}(c)$ auto-reject, and the rest defer. (If a refusal or crossed thresholds leave the auto-reject band empty, the overlap simply defers, and both guarantees still hold.)

G2 — certified false-reject, directly.. $t_{\mathrm{hi}}(c)$ is the split-conformal upper quantile of the *good* calibration scores at level α_{fr}: with $n_{\mathrm{cal}}^{\mathrm{good}}$ good calibration scores, it is the order statistic at rank $\lceil (n+1)(1-\alpha_{\mathrm{fr}}) \rceil$, guaranteeing $P(s \geq t_{\mathrm{hi}} \mid \mathrm{good}) \leq \alpha_{\mathrm{fr}}$ under exchangeability. This is exactly the split-conformal upper-quantile construction applied to the good scores.

G1 — certified escaped-defect, by negation.. Bounding $P(s \leq t_{\mathrm{lo}} \mid \mathrm{defective})$ is the mirror image: apply the same split-conformal construction to the *negated* defective calibration scores $Y = -X$ at level α_{miss}, then map the resulting upper quantile back through the negation. The gate module proves this order-statistic identity, and it is checked against the design's $\lfloor \alpha(n+1) \rfloor$ formula in the accompanying unit tests. No new estimator is introduced; both certificates are the same conformal quantile.

Mondrian per-category stratification.. Calibration is stratified by category (a Mondrian conformal construction): each category gets its own $(t_{\mathrm{lo}}, t_{\mathrm{hi}})$, so the guarantee is conditional on category, not marginal. An optional finer per-defect-type stratification is exploratory only (Section 6).

3.2. Certifiability floors and refusal

Split conformal cannot certify an arbitrarily small rate from a finite calibration pool: the smallest achievable level is $\alpha_{\min} = 1/(n_{\mathrm{cal}} + 1)$. A category is certifiable for a guarantee iff $\alpha_{\min} \leq \alpha$; otherwise the gate *refuses*, setting the corresponding threshold to $\pm\infty$ (the band is empty, all such images defer) and marking the category audited-not-certified for that axis. Refusal is emitted, never rounded away — this is the honesty rule that makes the floor table (Section 5.3) a result rather than a limitation.

3.3. Validity audit (V1) and kill-gates (K1–K7)

We fix $\alpha_{\mathrm{miss}} = 0.10$ and $\alpha_{\mathrm{fr}} = 0.05$ throughout, with a +3pp estimator tolerance (thresholds 0.13 / 0.08) and 95% one-sided Clopper–Pearson intervals [40]. The *validity audit* V1 has two tiers. Tier-1 checks that the realized mean escaped-defect and false-reject rates lie within target+3pp over repeated calibration/evaluation splits; this is the certificate-validity statistic. Tier-2 forms a Clopper–Pearson upper bound and is graded, per the frozen amendments, as a stringent estimator-variance readout (Section 5.10), because on MVTec AD its power floor is not met per repeat and its tolerance is near-unpassable by construction for a correctly-calibrated gate.

A battery of preregistered kill-gates guards the construction: K1 (coverage sanity — halt if tier-1 fails in ≥ 5 of the $15 \times B$ cells), K2 (vacuity — kill if median deferral > 0.80 in $\geq 8/15$ categories on all backbones), K3 (backbone floor — mean image-AUROC ≥ 0.90), K4 (audit headroom), K5 (calibration floor — fires only if $\alpha_{\min} > 0.10$ in ≥ 3 categories), K6 (a citation re-scan for a prior certified triage), and K7 (a compute guard).

3.4. Excess-AURC audit

Beyond validity, we ask whether standard thresholding practice earns deferral skill at all. For each practice — B1 (global fixed threshold), B2 (per-category tuned), B3 (train-good quantile) — we compare its area-under-the-risk-coverage-curve against the closed-form random-deferral null (evaluated in closed form, never Monte-Carlo'd), using a matched-abstention, category-stratified permutation p-value (2000 permutations), a category-blocked bootstrap CI, and a roster-derived Holm [41] correction. The verdict is symmetric: if no practice beats random the paper reports a debunking finding; if all do it reports a constructive one.

4. Experimental setup

Data.. Three public benchmarks. MVTec AD [1], staged from a fixed tarball (`sha256 = cf4313b1...98cd3d`): 15 object/texture categories, 3,629 train-good images, and a test set of 467 good + 1,258 defective = 1,725 images across 73 defect types. VisA [42], staged from the official `VisA_20220922` archive under the official `spot-diff` 1-class split: 12 categories, 8,659 train-good images, and a test set of 962 good + 1,200 anomalous = 2,162 images (VisA's public ground truth carries a single "bad" stratum per category, so the defect-type Mondrian axis is unavailable there). MPDD [29] (post-freeze exploratory; see below), 6 real painted-metal-part categories in native MVTec-AD directory layout, staged from the credential-free public HuggingFace mirror `meksamiao/mpdd` (`archive sha256 = 69f8da73...ddd6`, recorded in the frozen split manifest): 888 train-good images and a test set of 176 good + 282 defective = 458 images. Both box layouts (MVTec AD, VisA) were built by symlinking the official split file, refusing on any count mismatch; for MPDD, which ships without an external split file, the ground-truth test split is a frozen per-category manifest generated from the verified staged tree (the role the official one-class split file plays for VisA). The score dumps were adapted to the canonical schema with a per-image join against that split that refuses on any missing or extra image.

Backbones (Branch A, $B = 2$).. PatchCore [2] via anomalib [30] (a Wide ResNet-50-2 backbone, features from the second and third stages, coreset ratio 0.10)[1] and Dinomaly [3] (canonical score dump, its own eval script; on VisA, the reference repo's multi-class setting — one model per seed for all 12 categories — which is the setting its published VisA figure reports). Both were confirmed present and reproduction-passing before any calibration, so the preregistered two-backbone branch (Branch A) is active; the same two backbones and configurations are used unchanged on VisA.

Protocol.. Primary protocol: test-side good calibration, no train-holdout. For each (backbone, category, seed) we draw $R = 20$ stratified 50/50 calibration/evaluation splits (split-seed = repeat index), with 5 backbone seeds (0–4).

5. Results

All numbers in this section are recomputed from the frozen local score dumps by the archived analysis pipeline — a MVTec AD analysis stage, a stage-identical VisA analysis stage, a stage-identical MPDD analysis stage (with its own train-holdout rescue stage), and the G2-promotion stage, each emitting a frozen result summary; none is trusted from a training log or the literature.

5.1. Reproduction gate

On all three benchmarks the gate recomputes image-AUROC (Mann–Whitney rank-sum identity) directly from the local scores and compares it against a named published target (Table 1). On MVTec AD both backbones pass in all 5 seeds, and K3 (mean AUROC ≥ 0.90) holds with wide margin. On VisA, Dinomaly passes in all 5 seeds against its own published VisA figure (0.987, multi-class setting), and our recomputation again reproduces its reported per-seed mean image-AUROC seed for seed (max per-category deviation 3.3×10^{-5} over all 60 seed×category cells). PatchCore has no repo-confirmed published VisA figure, so per the tool's no-guessed-target rule its VisA row is *descriptive* (no pass/fail is graded); its recomputed mean of 0.903–0.908 still clears the K3 floor, though with thin margin (seed 3: 0.9033, i.e. 0.33pp above the 0.90 floor), and the drop from 0.982 (MVTec) is exactly the lower-ceiling substrate the audit needs (Section 5.7). On MPDD,

[1] Target 0.991 is PatchCore's published 25%-coreset mean; we score the 10%-coreset point (PREREG D3). See Table 1.

Dinomaly passes in all 5 seeds against its published MPDD multi-class figure (0.972; recomputed mean 0.9712–0.9738, grand mean 0.9722 ± 0.0010), our recomputation matching the box's per-category training-log AUROC to $\leq 4.9 \times 10^{-5}$; PatchCore-on-MPDD has no repo-confirmed published figure, so per the same no-guessed-target rule its row is descriptive (recomputed mean 0.948–0.957).

Table 1: Reproduction gate, 5 seeds per backbone per benchmark. Mean image-AUROC range is over seeds; the weakest category is named. "n/a": no repo-confirmed published PatchCore-on-VisA/MPDD figure exists, so that row is descriptive by design (no pass/fail graded). Recomputed from the frozen MVTec AD, VisA, and MPDD analysis summaries.

Benchmark	Backbone	Target (± 0.02)	Mean I-AUROC (seed range)	Weakest category	Gate
MVTec AD	PatchCore	0.991	0.9817–0.9826	toothbrush 0.911–0.917	PASS 5/5
MVTec AD	Dinomaly	0.996	0.9957–0.9965	capsule 0.976–0.982	PASS 5/5
VisA	PatchCore	n/a	0.9033–0.9083	macaroni2 0.656–0.665	descriptive
VisA	Dinomaly	0.987	0.9866–0.9876	macaroni2 0.956–0.966	PASS 5/5
MPDD	PatchCore	n/a	0.9481–0.9568	bracket black 0.872–0.888	descriptive
MPDD	Dinomaly	0.972	0.9712–0.9738	bracket black 0.929–0.940	PASS 5/5

5.2. Cross-seed stability

PatchCore's five seeds are genuinely distinct: all 60 seed-0-vs-seed-N score-table comparisons (15 categories $\times$ 4 seeds) are 0/60 bit-identical, with max $|\Delta\text{score}|$ ranging 0.026–0.153 (mean 0.074) — a real but modest variance source, so PatchCore is reported as a 5-effective-seed backbone, not collapsed to one. Dinomaly is low-variance: mean image-AUROC across seeds 0.9960 ± 0.00029, with the widest per-category spread only 0.0060 (capsule). The same picture holds on VisA: PatchCore is 0/48 bit-identical (max $|\Delta\text{score}|$ 0.044–0.256), and Dinomaly's mean image-AUROC across seeds is 0.9870 ± 0.0004 with the widest per-category spread 0.0104 (macaroni2). On MPDD the pattern repeats once more: PatchCore is 0/24 bit-identical across the seed-0-vs-seed-N comparisons, and Dinomaly's mean image-AUROC across seeds is 0.9722 ± 0.0010.

5.3. Certifiability floors and the G1/G2 split

Table 2 is the paper's honesty figure. The per-category calibration counts computed from the full local test arrays match the preregistered arithmetic-only floor table exactly (0/15 mismatches for both backbones, identical across all 5 seeds). At the frozen rates, escaped-defect (G1) is certifiable in all 15 categories; false-reject (G2) is certifiable in exactly 4 — cable, hazelnut, screw, transistor — the categories with $n_{\text{cal}}^{\text{good}} \geq 19$. The other 11 categories are refused for G2 and reported audited-not-certified. We are careful about what this exact match does and does not show: the per-category counts are a *deterministic* function of the fixed MVTec test split (round-half-to-even of $n/2$), so the 0/15 agreement with the preregistered table is a pipeline-correctness check — it confirms the data staging re-derives the frozen counts with no off-by-one or CSV-read error — not an empirical result about the gate. The genuinely empirical result is the V1 tier-1 validity audit below (Section 5.6), measured from real calibrate/route/evaluate runs, which could have failed had the calibration or routing pipeline been broken.

Table 2: Per-category certifiability at $\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$ (backbone-invariant; pure functions of the shared MVTec test split). G1 certifies in 15/15, G2 in 4/15. Source: the frozen preregistration (§3), reproduced with 0/15 mismatch by the analysis pipeline.

Category	$n_{\text{cal}}^{\text{def}}$	$\alpha_{\min}$(G1)	G1@0.10	$n_{\text{cal}}^{\text{good}}$	G2@0.05
bottle	32	0.0303	OK	10	REFUSE
cable	46	0.0213	OK	29	OK
capsule	54	0.0182	OK	12	REFUSE
carpet	44	0.0222	OK	14	REFUSE
grid	28	0.0345	OK	10	REFUSE
hazelnut	35	0.0278	OK	20	OK
leather	46	0.0213	OK	16	REFUSE
metal_nut	46	0.0213	OK	11	REFUSE
pill	70	0.0141	OK	13	REFUSE
screw	60	0.0164	OK	20	OK
tile	42	0.0233	OK	16	REFUSE
toothbrush	15	0.0625	OK	6	REFUSE
transistor	20	0.0476	OK	30	OK
wood	30	0.0323	OK	10	REFUSE
zipper	60	0.0164	OK	16	REFUSE
Certifiable			15/15		4/15

VisA: the same arithmetic, a friendlier split.. On VisA the identical floor computation lands on the opposite side of the ledger: every category's test split carries 100 anomalous and 50–101 good images, so after the 50/50 calibration split $n_{\text{cal}}^{\text{def}} = 50$ everywhere ($\alpha_{\min}^{\text{G1}} = 0.0196$) and $n_{\text{cal}}^{\text{good}} \in \{25, 30, 50\}$ ($\alpha_{\min}^{\text{G2}} =$ 0.0196–0.0385), all below $\alpha_{\text{fr}} = 0.05$. G1 and G2 both certify in 12/12 categories under the primary protocol alone — no train-holdout remedy is needed on VisA — and the floors are again identical across both backbones and all five seeds (0 mismatches). This is the refusal mechanism working in both directions: it refuses 11/15 categories on MVTec AD because the data cannot support the rate, and certifies 12/12 on VisA because it can.

MPDD: the stingy extreme of pool-size-driven refusal.. MPDD's per-category test-good pools are uniformly small (26–32 images $\rightarrow n_{\text{cal}}^{\text{good}}$ 13–16, every one below the 19 the false-reject floor needs at $\alpha_{\text{fr}} = 0.05$), so G2 is certifiable in 0/6 categories under the primary protocol, on both backbones, identical across all five seeds (0 mismatches). G1 (escaped-defect) stays healthy at 5/6 — only connector fails, its 14 test-defective yielding $n_{\text{cal}}^{\text{def}} = 7 < 9$. This places MPDD at the stingy end of three points consistent with pool-size-driven refusal, ordered by good-pool size: MPDD 0/6 $\rightarrow$ MVTec AD 4/15 $\rightarrow$ VisA 12/12 (all three computed apples-to-apples by the identical floor code path). MPDD is thus not an intermediate point but the cleanest demonstration of the paper's thesis — the certifiable count is a property of how much exchangeable good data the protocol can spend, and the gate refuses precisely when the data cannot pay for a 5%-false-reject certificate. Figure 2 makes this certify-or-refuse pattern visible at the category level across all three benchmarks: VisA certifies both axes in every category, MVTec AD's false-reject row is hatched (refused) everywhere except the four categories in Table 2, and MPDD's false-reject row is hatched throughout — the visual complement of the same table.

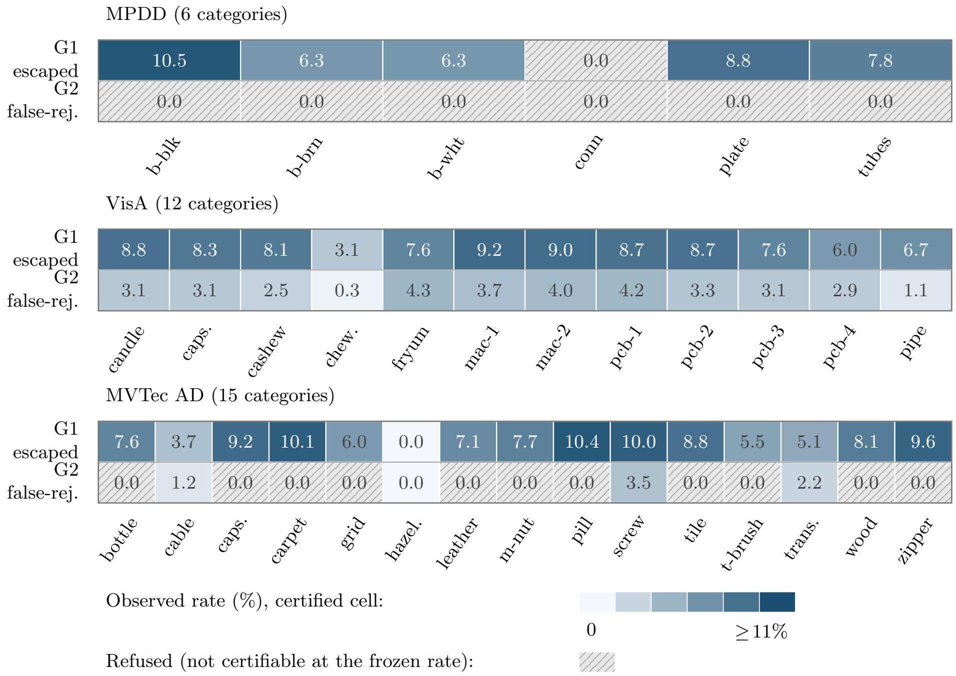

Figure 2: Per-category certification map for escaped-defect (G1) and false-reject (G2) at the frozen rates ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$), pooled over both backbones and five seeds, for MPDD, VisA, and MVTec AD. Colored cells are certified, shaded by the observed rate; hatched gray cells are refused as not certifiable at the frozen rate under the primary protocol (all of MPDD's G2 column, MPDD's connector on G1, and the 11/15 MVTec AD categories below the false-reject floor in Table 2).

5.4. G2 train-holdout promotion on MVTec AD (frozen remedy arm)

The preregistered calibration-efficiency remedy — recalibrating G2 on held-out *train*-good scores, gated behind a per-category KS exchangeability test with a mandatory audited-not-certified fallback — was computed on 2026-07-12 under the 2026-07-11 freeze, for PatchCore only: Dinomaly has no train-side score dump (its reference eval scores test images only), so the promotion arm is single-backbone by construction, disclosed here rather than implied otherwise. It behaves as preregistered: G2-certifiable categories rise from 4/15 to 13/15 in seeds 0–3 and 12/15 in seed 4. The KS gate does real work rather than rubber-stamping: leather fails it in all five seeds (its train-good and test-good score distributions are not exchangeable) and screw additionally fails in seed 4 — each is reported audited-not-certified, never silently promoted, and the seed-4 screw case shows a category certified under the primary protocol can still be *refused* under the holdout arm when exchangeability breaks. Toothbrush remains uncertifiable at $\alpha_{\mathrm{fr}} = 0.05$ under both arms, exactly as preregistered (n^{good} too small in either pool). K1 is not tripped in any promoted seed aggregate.

5.5. MPDD train-holdout rescue (post-freeze exploratory)

We report the primary-protocol 0/6 of the preceding section as MPDD's thesis-confirming headline; the train-holdout arm below is a clearly secondary, post-freeze, PatchCore-only result. The train-holdout calibration-efficiency lever the frozen MVTec remedy exercises (Section 5.4) turns MPDD's 0/6 primary refusal into a 4/6 rescue, a vivid single-benchmark illustration of the same lever. MPDD's train-good pools are large (54–289 per category) where its test-good pools are tiny, so recalibrating G2 on a 20% held-out train-good pool (the flag-gated train-holdout calibration mode, 20% holdout fraction; PatchCore only, as on MVTec — Dinomaly dumps no train-side scores, so the arm is single-backbone by construction) clears the false-reject floor for the 5/6 categories whose holdout pool reaches $n \geq 19$. The same per-category KS exchangeability gate then adjudicates each promotion, and it does decisive work: G2-certifiable rises from 0/6 to 4/6, identical in all five seeds. Two categories are honestly held back — metal plate, whose holdout pool (11) is still below the floor, and tubes, which fails the KS gate (BH-adjusted $p \approx 1.2 \times 10^{-5}$; its train-good and test-good score distributions are demonstrably non-exchangeable) and is reported audited-not-certified rather than promoted. The remaining four (bracket black, bracket brown, bracket white, connector) promote cleanly (KS $p_{\mathrm{BH}} \geq 0.44$ every seed). Connector is the one category G1 refuses under the primary protocol ($n^{\mathrm{def}}_{\mathrm{cal}} = 7 < 9$ defective) yet the arm G2-certifies (holdout-good $26 \geq 19$): the two certificates gate on independent sample axes — defect count for escaped-defect, good count for false-reject — so this is coherent, not contradictory. Its 100% deferral (the highest of any category, since G1 refusal leaves the auto-reject band empty) nonetheless makes that G2 certificate operationally hollow in practice. So the count-only floor of 5/6 is audited down to a genuinely certified 4/6 — the KS guardrail keeping the rescue honest, exactly the exchangeability risk MPDD's non-homogeneous backgrounds were expected to pose. As a sanity check (no gate), the holdout arm's 80%-memory-bank test scores rank-correlate with the primary 100%-bank scores at Spearman $\rho \geq 0.918$ across all 6×5 category-seed cells (max $|\Delta\text{image-AUROC}| = 0.040$), confirming the same model was scored. This arm is post-freeze exploratory and flag-gated prereg-neutral (it re-freezes nothing and is never in the confirmatory Holm family); the $0/6 \to 4/6$ contrast on the same benchmark and the same scores is a vivid single-benchmark demonstration of the calibration-efficiency lever.

The larger, *confirmatory* instance of the same lever is the frozen MVTec train-holdout remedy (Section 5.4), which lifts G2 from 4/15 to 12–13/15 across seeds (13 in seeds 0–3, 12 in seed 4). The primary-protocol 0/6 stays the honest headline for this industrial benchmark: on real metal parts the gate certifies nothing at the 5% false-reject budget because the data cannot pay for it — exactly what a refusal-first design should do.

5.6. Validity audit V1 (tier-1) and the kill-gates

Per frozen amendment A3, tier-1 is reported *per axis*, with the false-reject axis evaluated only over G2-certified cells: wherever G2 is refused the auto-reject region is empty and the measured false-reject rate is identically 0 — a pass without evidence that A3 excludes from every count. On the escaped-defect axis, tier-1 passes in all 150 MVTec cells (2 backbones $\times 5$ seeds $\times 15$ categories), all 120 VisA cells ($2 \times 5 \times 12$), and all 60 MPDD cells ($2 \times 5 \times 6$), every rate within target+3pp. On the false-reject axis, MVTec has 40 evaluable cells (4 G2-certified categories $\times 2 \times 5$) and passes 40/40, with the other 110 cells reported G2-REFUSED and excluded; on VisA every category is G2-certified, so all 120/120 cells are evaluable and pass non-vacuously; on MPDD every category is G2-refused under the primary protocol, so all 60 false-reject cells are G2-REFUSED and excluded (the mirror image of VisA, and the primary-protocol counterpart to the train-holdout rescue of Section 5.5). Tier-1 is a necessary-not-sufficient pipeline check — a conservative gate realizes rates well below target, so passing validates the calibrate/route/evaluate machinery rather than demonstrating sharpness. Consequently K1 is not tripped (the A3 exclusion cannot change K1's count: excluded cells cannot fail) (0 violations in every (backbone, seed) aggregate, against the 5-cell MVTec/VisA kill threshold and, on MPDD, against a proportionally-scaled 2-cell threshold[2]) and K2 is not tripped (on MVTec/VisA at most 1 vacuous category — VisA/PatchCore/seed 2 — against the 8-category/0.80-deferral threshold; on MPDD at most 2 vacuous categories per aggregate against the scaled 4-category threshold), in all 30 aggregates across the three benchmarks (Table 3). Median per-category deferral at seed 0 on MVTec AD is 70.0% for PatchCore (mean 55.1%; max pill 78.0%, min transistor 2.0%) and 69.5% for Dinomaly (mean 54.2%; max capsule 77.3%, min screw 3.1%) — elevated because G2 refusal empties the auto-reject band in 11/15 categories, but far from the vacuity threshold.

[2]Absolute kill thresholds are un-trippable at $n_{\text{cat}} = 6$, so MPDD uses proportional scales (K1$\rightarrow$ 2, K2$\rightarrow$ 4); diagnostic only.

On VisA, where G2 certifies everywhere and the auto-reject band is live in every category, seed-0 median deferral drops to 18.7% for PatchCore (max macaroni2 72.5%, min chewinggum 2.0%) and 4.2% for Dinomaly (max capsules 21.9%, min candle 2.5%) — direct operational evidence that the elevated MVTec deferral was the price of honest G2 refusal, not a property of the gate. MPDD sits at the far end again: with G2 refused in all 6 categories under the primary protocol, seed-0 median deferral is 74.1% for PatchCore (max connector 100%, min bracket brown 61.8%) and 70.9% for Dinomaly (max connector 100%, min bracket white 53.3%), the highest of the three benchmarks and exactly what 0/6 certification predicts. These are three honest operating points, not a defect and a fix: MPDD's $\sim 72\%$ and the 55–70% MVTec deferral are the cost of refusing to certify small good-calibration pools, and VisA's 4–19% is what the same gate, same code path, spends when the good pools are large enough to certify every category.

Table 3: V1 tier-1 (per axis, per frozen amendment A3) and kill-gates, aggregated over the 30 (backbone, seed) cells across all three benchmarks. †False-reject evaluated only over the 4 G2-certified MVTec categories; the other 110 cells are G2-REFUSED and excluded from the count, never counted as (vacuous) passes. ‡On MPDD all 60 false-reject cells are G2-REFUSED under the primary protocol and excluded (see Section 5.5 for the train-holdout rescue). MPDD's K1/K2 thresholds are proportionally scaled to $n_{\text{cat}} = 6$ (2 and 4). "—" marks a cell that does not apply to that row (the raw evaluable-cell counts above have no separate threshold or trip verdict; K5 never fires under the frozen design, so it reports no single value). Sources: the frozen MVTec AD, VisA, and MPDD analysis summaries and their per-cell validity dumps; frozen preregistration §9 amendment A3.

Axis	Value	Threshold	Tripped?
V1 tier-1 escaped-defect cells (MVTec AD)	150/150	—	—
V1 tier-1 false-reject cells (MVTec AD)†	40/40	—	—
V1 tier-1 escaped-defect cells (VisA)	120/120	—	—
V1 tier-1 false-reject cells (VisA, all G2-certified)	120/120	—	—
V1 tier-1 escaped-defect cells (MPDD)	60/60	—	—
V1 tier-1 false-reject cells (MPDD)‡	0 evaluable	—	—
K1 coverage violations (per cell)	0	≥ 5 (≥ 2 MPDD)	No (all 30)
K2 vacuous categories (per cell)	0 (1 in VisA/ PatchCore/seed 2; ≤ 2 on MPDD)	≥ 8 (≥ 4 MPDD)	No (all 30)
K3 backbone floor (mean AUROC)	≥ 0.982 (MVTec) / ≥ 0.903 (VisA) / ≥ 0.948 (MPDD)	< 0.90	No
K5 calibration floor	—	$\alpha_{\min} > 0.10$ (≥ 3 cat)	Cannot fire

5.7. *Exploratory excess-AURC audit*

As an exploratory (not confirmatory) readout scoped to one (backbone, seed, category) family of size 2 (practices B1 and B2; B3 could not run — see below), on the repeat-0 split at seed 0: on MVTec AD, PatchCore rejects the random-deferral null in 10/30 practice-cells after within-cell Holm, with 6/30 degenerate (per-category AUROC $= 1.0 \Rightarrow$ zero measurable headroom); Dinomaly rejects in 0/30 with 14/30 degenerate — an expected ceiling effect, since Dinomaly sits at per-category AUROC 1.0 in 8/15 categories on this split, leaving threshold practices no room to show skill above a near-perfect baseline. VisA resolves this ceiling effect: on the same seed-0 readout, PatchCore rejects in 14/24 cells and Dinomaly in 16/24, with 0 degenerate cells for either backbone — on a benchmark where the backbones are not saturated, field-standard threshold practices demonstrably do earn deferral skill over honest random deferral, so the audit's verdict axis is live rather than vacuously constructive. Across all five seeds the picture is stable (MVTec: PatchCore 48/150, Dinomaly 4/150; VisA: PatchCore 64/120, Dinomaly 42/120 practice-cell rejections). One comparability caveat: on VisA, Dinomaly is a multi-class model (one model per seed for all 12 categories) while PatchCore keeps per-category memory banks, so cross-backbone audit comparisons there are not architecture-controlled and are read per backbone, not head to head.

5.8. Confirmatory audit verdict (C2)

The exploratory readout above pools nothing; its higher-powered confirmatory counterpart pools the 15 MVTec categories per (practice, backbone). The confirmatory family is the four tests $\{\text{B1 fixed}, \text{B2 tuned}\} \times \{\text{PatchCore}, \text{Dinomaly}\}$; the train-good quantile practice B3 drops out because the frozen confirmatory pass loaded no train-good scores, the degradation from six to four tests preregistered in PREREG §4. B3 is not infeasible in general: a held-out train-good score pool does exist for PatchCore, so its B3 arm is completed post-hoc below; Dinomaly, however, has no train-side score dump at all, so the full six-member family cannot be run even now. For each (practice, backbone) we pool the 15 categories into one evaluation half (repeat-0 split), fit B1 on the pooled calibration half and B2 per category, and test the excess area under the risk–coverage curve against the closed-form random-deferral null with a matched-abstention permutation test ($n_{\text{perm}} = 2000$, strata = category) and a category-blocked bootstrap CI. Every one of the four tests rejects the random-deferral null at the permutation floor $p = 1/2001 = 5 \times 10^{-4}$ (Holm-adjusted $p = 2 \times 10^{-3}$) in all five backbone seeds, with pooled excess-AURC in $[0.023, 0.050]$ and every bootstrap CI excluding zero (Table 4). The frozen confirmatory construction is the per-seed four-member Holm family, and it rejects in every one of the five seeds. The preregistration is silent on how to combine seeds, so we report the seed dimension as a robustness check rather than a family axis: the single cross-seed rollup (the seed-max of the per-seed p-values) was authored post-freeze as a one-shot convenience and is moot here, the verdict being identical under seed-min, seed-median, and seed-max. The constructive arm of C2 publishes: field-standard threshold practice earns real deferral skill over honest random deferral. Pooling recovers across the 15 categories the power that per-category saturation denies the exploratory single-category readout.

B3 (train-good quantile), the family's feasible half, post-hoc.. Because a held-out train-good score pool exists for PatchCore (the 2026-07-10 holdout run scored the per-category train-good images alongside the test set), we complete the third practice for that backbone post-hoc, on the same pooled-category construction. B3 is constructive and consistent with B1/B2: across all five seeds it rejects the random-deferral null at the permutation floor $p = 5\times10^{-4}$, with pooled excess-AURC in $[0.053, 0.056]$ (larger than B1's and B2's) and every category-blocked bootstrap CI excluding zero; a cross-run sensitivity arm agrees. Added to the frozen four-member per-seed family as a fifth member, all five reject in all five seeds (Holm $p = 2.5 \times 10^{-3}$). This is labelled post-hoc and exploratory: the confirmatory verdict remains the frozen four-member family above, and B3-Dinomaly stays impossible for want of a train-side dump.

On VisA (post-freeze, exploratory, and kept out of the MVTec confirmatory Holm family) the same pooled construction rejects the random-deferral null in all four practice×backbone tests across all five seeds (Holm $p = 2\times10^{-3}$), with larger pooled excess-AURC $[0.044, 0.106]$ — the headroom the near-saturated MVTec backbones deny (Table 4, lower block).

Table 4: Pooled excess-AURC audit (C2). Family = {B1 fixed, B2 tuned} × {PatchCore, Dinomaly}, Holm $\alpha = 0.05$. MVTec AD is the frozen confirmatory family: all four tests reject the random-deferral null in every one of the five backbone seeds, and the per-seed values are the frozen confirmatory verdict. VisA (post-freeze) is reported exploratory and is kept out of the MVTec Holm family. Excess-AURC is the seed range; permutation floor $p = 1/2001$.

Benchmark	Backbone	Practice	excess-AURC (seed range)	perm. p	Holm p / reject
MVTec AD	PatchCore	B1 fixed	0.023–0.030	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	PatchCore	B2 tuned	0.036–0.046	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	Dinomaly	B1 fixed	0.045–0.050	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	Dinomaly	B2 tuned	0.027–0.035	5×10^{-4}	2×10^{-3} / ✓
VisA	PatchCore	B1 fixed	0.073–0.089	5×10^{-4}	2×10^{-3} / ✓
VisA	PatchCore	B2 tuned	0.086–0.106	5×10^{-4}	2×10^{-3} / ✓
VisA	Dinomaly	B1 fixed	0.081–0.095	5×10^{-4}	2×10^{-3} / ✓
VisA	Dinomaly	B2 tuned	0.044–0.055	5×10^{-4}	2×10^{-3} / ✓

Verdict: constructive arm on both benchmarks. All four MVTec tests reject in all five seeds (frozen confirmatory); all four VisA tests reject in all five seeds (post-freeze exploratory). Every bootstrap CI excludes 0.

5.9. *When a fixed threshold over-promises (post-hoc binding demonstration)*

The excess-AURC audit shows practice earns deferral skill, but it does not directly exhibit the sharper event: a category where a *naive fixed threshold* actually violates a target error rate while the certified gate holds. We test that event directly, post-hoc. For each (backbone, category) we take the naive practitioner's operating point — a single global best-F1 threshold (B1) applied to every eval image with no deferral — and compute its realized escaped-defect and false-reject rates on the held-out eval half; alongside it we compute the certified gate's realized rates and its deferral cost on the same split (mean over the $R = 20$ repeats, at backbone seed 0). A cell *binds* when the fixed threshold's realized rate exceeds the target ($\alpha_{\text{miss}} = 0.10$ or $\alpha_{\text{fr}} = 0.05$) while the gate's stays within target+3pp.

Such cells exist, on both benchmarks, both axes, and both backbones. Restricting to the strongest evidence — cells where the gate is certified on that axis in all 20 repeats and the bind holds in all five backbone seeds — there are 27: on MVTec AD, 3 escaped-axis (Dinomaly on capsule, pill, screw) and 3 false-reject-axis (PatchCore on screw; Dinomaly on cable, transistor); on VisA, 4 escaped-axis and 17 false-reject-axis cells. Crucially the effect is not an artifact of the gate abstaining on everything: the cleanest cases pay a small deferral cost. On MVTec AD a global threshold lets 24.2% of defective screws escape; the certified gate, deferring just 4% of images, holds the escaped rate at 8.0% (under target) because its per-category (Mondrian) calibration puts the auto-pass boundary where a global threshold cannot. On the false-reject axis, the same global threshold auto-rejects 19.5% of good screws while the gate holds false-reject at 5.0% (certified, 14% deferral); on VisA the extreme case is capsules, where a global threshold would auto-reject 96.8% of good parts and the gate holds false-reject at 3.7%. The mechanism the demonstration exposes is that a single fixed *global* threshold cannot serve categories with different score scales at once, so it over-promises on whichever it mis-fits; the certified gate refuses (defers) exactly where it cannot honor the target and calibrates per category elsewhere. This is a post-hoc, exploratory illustration — not a preregistered arm — and is reported verdict-honestly: had no cell bound, we would have kept the "not demonstrated" framing.

5.10. *V1 tier-2 verdict*

Under the frozen amendments the tier-2 power floor is applied per repeat, not pooled over repeats (amendment A1). On MVTec AD the escaped-defect axis is per-repeat powered in 13/15 categories (underpowered in toothbrush and transistor, per-repeat $n_{\text{eval,def}} = 15, 20 < 22$). Graded on a per-repeat one-sided 95% Clopper–Pearson upper bound against the 0.13 threshold, the powered cells overwhelmingly fail (115 fail / 15 pass out of 130 powered, 20 excluded as underpowered: toothbrush, transistor; the 15 passes are the near-zero-miss cells — Dinomaly on cable and hazelnut, PatchCore on hazelnut). This is the preregistered expectation of amendment A2, not a certificate failure: split-conformal calibration targets α_{miss} exactly, so a one-sided upper bound at a per-repeat $n \approx 30$–70 sits above the target-plus-3pp tolerance whenever the realized miss rate is near the target. The certificate-validity statistic is tier-1, which passes in all 150 MVTec cells; tier-2 is a stringent estimator-variance readout layered on top (Table 5 gives the per-category breakdown). The tier-2 false-reject axis is *structurally ungraded* on MVTec AD: it is scored only over G2-certified categories (amendment A3), and those four (cable, hazelnut, screw, transistor) are themselves per-repeat underpowered ($n_{\text{eval,good}} \leq 30 < 36$), so 0 pass / 0 fail out of 150 cells are gradeable (110 G2-refused $+\,40$ underpowered excluded) — exactly the amendment A1+A3 outcome. One caveat we disclose: the power partition and the false-reject buckets are exact (deterministic from the cached counts), but the escaped pass/fail split within powered cells is reconstructed from a per-repeat Clopper–Pearson upper bound built on the pooled rate as the per-repeat point estimate, not from the raw per-repeat records. On VisA (post-freeze, exploratory) the larger good pools clear the false-reject power floor in part, so that axis is partially gradeable there: escaped 17 pass / 103 fail (of 120), false-reject 4 pass / 66 fail / 50 underpowered (of 120).

Table 5: V1 tier-2 grading, per category (expanded from the aggregate axis totals in the text; every cell out of $10 = 2$ backbones $\times$ 5 seeds). Escaped-defect is pass/fail/unpowered against the per-repeat 0.13 Clopper–Pearson bound (amendment A2); false-reject is pass/fail/unpowered/refused, where "refused" means the category is not G2-certified (amendment A3) and "unpowered" means it is G2-certified but fails the per-repeat power floor (amendment A1). MVTec AD is 0-gradeable on false-reject by construction (no G2-certified category clears the power floor); VisA (post-freeze, exploratory) is partially gradeable.

Category	Escaped-defect (pass/fail/unpwr.)	False-reject (pass/fail/unpwr./refused)
MVTec AD (confirmatory tier-1 family; amendments A1/A2/A3)		
bottle	0/10/0	0/0/0/10
cable	5/5/0	0/0/10/0
capsule	0/10/0	0/0/0/10
carpet	0/10/0	0/0/0/10
grid	0/10/0	0/0/0/10
hazelnut	10/0/0	0/0/10/0
leather	0/10/0	0/0/0/10
metal_nut	0/10/0	0/0/0/10
pill	0/10/0	0/0/0/10
screw	0/10/0	0/0/10/0
tile	0/10/0	0/0/0/10
toothbrush	0/0/10	0/0/0/10
transistor	0/0/10	0/0/10/0
wood	0/10/0	0/0/0/10
zipper	0/10/0	0/0/0/10
VisA (post-freeze, exploratory)		
candle	0/10/0	0/10/0/0
capsules	0/10/0	0/0/10/0
cashew	0/10/0	0/0/10/0
chewinggum	10/0/0	0/0/10/0
fryum	0/10/0	0/0/10/0
macaroni1	0/10/0	0/10/0/0
macaroni2	0/10/0	0/10/0/0
pcb1	0/10/0	0/10/0/0
pcb2	0/10/0	0/10/0/0
pcb3	0/10/0	0/10/0/0
pcb4	5/5/0	4/6/0/0
pipe_fryum	2/8/0	0/0/10/0
Totals		
MVTec AD (150 cells)	15/115/20	0/0/40/110
VisA (120 cells)	17/103/0	4/66/50/0

5.11. Comparison with a conformal single-threshold baseline

The natural named baseline a reviewer would ask for is not a hand-picked straw man but the single-threshold construction our own G1 certificate specializes: split-conformal Conformal Risk Control (CRC) [36], applied with one per-category threshold at $\alpha_{\text{miss}} = 0.10$ and no defer option. For the 0/1 miss loss this threshold is exactly our G1 split-conformal threshold (an identity proved in the gate module's implementation), so CRC and our gate share the same finite-sample escaped-defect guarantee and certify the same cells; the only degree of freedom is what each does with the ambiguous middle. We recompute CRC on the identical canonical score dumps, under the identical protocol ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$, $R = 20$ repeated stratified 50/50 calibration/evaluation splits, five seeds, both backbones), so every cell in Table 6 is apples-to-apples with the main results. CRC has no mechanism to certify the false-reject side at all — with one threshold, every non-passed image is an auto-reject, not a deferral — so its G2 column is structurally n/a.

Table 6: Our dual gate vs. a single-threshold CRC baseline controlling the same escaped-defect risk ($\alpha_{\text{miss}} = 0.10$), pooled over both backbones, five seeds, and all categories ($n_{\text{cells}} = 2 \times 5 \times n_{\text{categories}}$ per benchmark). CRC certifies the same escaped-defect cells (G1) but has no false-reject certificate (G2 n/a) and must auto-reject the entire ambiguous band instead of deferring it.

Benchmark	Method	G1 cert.	G2 cert.	Escaped	False-reject	Deferral
MPDD	Our dual gate	50/60	0/60	6.6%	0.0%	73.1%
MPDD	CRC (single thr.)	50/60	n/a	6.6%	36.9%	0.0%
VisA	Our dual gate	120/120	120/120	7.6%	3.0%	16.4%
VisA	CRC (single thr.)	120/120	n/a	9.5%	16.2%	0.0%
MVTec AD	Our dual gate	150/150	40/150	7.3%	0.5%	54.4%
MVTec AD	CRC (single thr.)	150/150	n/a	8.5%	3.1%	0.0%

On MPDD, where the false-reject floor refuses G2 entirely (Section 5.3), the contrast is starkest: CRC pays 36.9% false-reject to hold the same 6.6% escaped-defect rate our gate holds at 0.0% false-reject, by deferring 73.1% of images instead of auto-rejecting them. VisA and MVTec AD show the same direction at smaller magnitude (16.2% vs. 3.0% false-reject on VisA; 3.1% vs. 0.5% on MVTec AD) — CRC's realized escaped-defect rate is also somewhat higher on these two benchmarks (9.5% vs. 7.6% on VisA, 8.5% vs. 7.3% on MVTec AD), consistent with both methods controlling the same risk to within realized-rate sampling noise across repeats rather than producing bit-identical per-repeat routes. The comparison isolates exactly what the dual-gate design buys over the nearest published single-threshold alternative: not a better escaped-defect guarantee (both share the same conformal construction) but a false-reject guarantee CRC cannot offer, purchased by turning auto-rejects into deferrals. Figure 3 plots the three rates for both methods side by side across all three benchmarks, making the MPDD contrast in particular immediately visible.

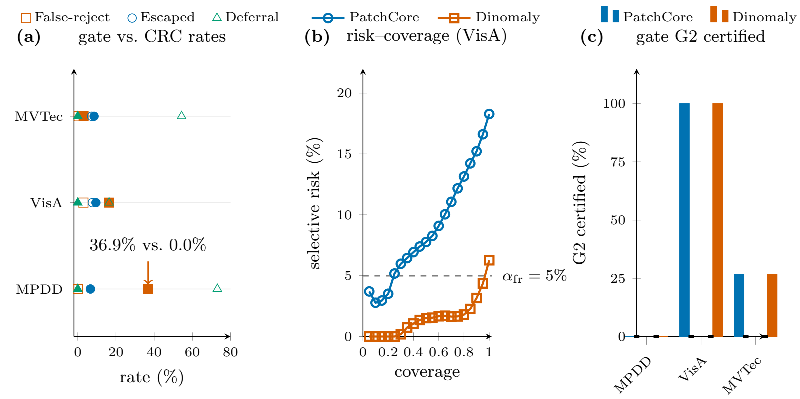

Figure 3: Escaped-defect, false-reject, and deferral rates for our dual gate versus the single-threshold CRC baseline (Table 6), pooled over both backbones and five seeds. CRC has no deferral mechanism, so every non-passed image becomes an auto-reject; on MPDD this drives its false-reject rate to 36.9% against 0.0% for our gate at the same escaped-defect rate.

As a second, non-conformal reference line we also compute the field-standard selective-prediction risk-coverage curve [31] (per-category best-F1 threshold, margin-based abstention) on the same score dumps. Its area-under-the-risk-coverage-curve (AURC, lower is better) ranges from 0.0019 (MVTec AD, Dinomaly) to 0.0849 (VisA, PatchCore); at $\sim 80\%$ coverage the realized risk is 0.0625/0.0154 (MPDD, PatchCore/Dinomaly), 0.1315/0.0180 (VisA, PatchCore/Dinomaly), and 0.0095/0.0026 (MVTec AD, PatchCore/Dinomaly). This is a fundamentally different guarantee — a curve traced out by sweeping coverage post hoc, not a fixed pre-specified certificate at a chosen risk level — so we report it as context rather than a like-for-like comparison to Table 6.

5.12. Latency and deployment cost

The gate is a thin decision layer on top of whatever detector already produces the anomaly score, so the only cost it adds is the conformal calibration (once, per calibration refresh) and the per-image routing comparison. We measured both on this work's hardware, and we place them alongside the backbone forward pass that dominates the end-to-end budget (Table 7).

Routing an image is an $O(1)$ threshold comparison inside an $O(\text{sort})$ calibration; the measured per-image routing cost is $2.8\,\mu$s on CPU — roughly four orders of magnitude below the backbone forward pass and under 0.02% of Dinomaly's per-image GPU inference. Calibrating the full three-way gate over one 861-image calibration half across all 15 MVTec strata (the $\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$ primary configuration) takes 2.4 ms end to end, and is amortized over an entire evaluation run rather than paid per image. The practical conclusion is that certification is free relative to detection: a deployment already able to afford the detector can add the escaped-defect / false-reject certificate at no measurable latency cost. The backbone rows are the honest budget the gate sits on — Dinomaly measured here on an RTX 5090 Laptop GPU, PatchCore's row taken from the authors' reported figure (not measured here, and both hardware- and coreset-dependent). We flag the substrate mix these numbers span, so no cross-row comparison is over-read: the gate rows are timed on CPU over the MVTec calibration scores (861-image cal half, 15 strata), the Dinomaly backbone row on GPU over 250 real VisA test images (the VisA branch-A seed-0 checkpoint), and the PatchCore row is the authors' published $\sim$1%-coreset figure whereas our scoring uses the 10%-coreset configuration — so the table brackets the order of magnitude of the backbone budget the gate rides on, and it is the μs-scale CPU gate overhead, not any CPU-vs-GPU or MVTec-vs-VisA backbone comparison, that the paper contributes.

Table 7: Per-image latency and deployment cost. The two gate rows and the Dinomaly backbone row are *measured on this work's hardware* (gate: Intel Core Ultra 9 275HX CPU, 24 threads; Dinomaly: NVIDIA RTX 5090 Laptop GPU, torch 2.12.0, batch 1, median over 250 real VisA test images, full forward + anomaly-map + image-score pass). The PatchCore backbone row is *reported by the authors* (not measured here; Roth et al. [2], ∼1%-coreset configuration on their hardware) and is included only to bracket the backbone budget — backbone timing is hardware- and configuration-dependent, whereas the gate overhead below it is the measured constant this paper contributes. Sources: the frozen latency measurement dumps.

Component	Per-image cost	Notes
Backbone forward pass (dominates the budget)		
Dinomaly (DINOv2-b, 148M) — measured, GPU	18.9 ms (p95 23.1)	53 img/s at batch 1; 17.0 ms/img batched (B=32)
PatchCore (WRN50, coreset) — *authors' figure*	$\lesssim$ 200 ms	Roth et al. [2], their hardware; not measured here
Gate overhead added on top (this paper) — measured, CPU		
Routing (per image)	2.8 μs	$O(1)$ compare; < 0.02% of the Dinomaly forward pass
Calibration (one-time)	2.4 ms (p95 2.8)	full 3-way gate, 861-img cal half, 15 strata; per refresh

6. Limitations and gated arms

The preregistration was frozen on 2026-07-11 (a sha256-pinned sign-off record over the preregistration document and its A1–A3 amendments). One frozen arm has since been computed and is reported above (the G2 train-holdout promotion, Section 5.4). The VisA and MPDD benchmarks are both post-freeze additions — the frozen preregistration mentions neither — run under the identical frozen protocol and code path but reported as exploratory, never confirmatory, throughout Section 5; MPDD's train-holdout rescue (Section 5.5) is additionally a flag-gated, prereg-neutral protocol variant. The following arms remain *unlocked but not yet computed*; each is shown as an explicit slot and is not reported as a result anywhere above.

- **Confirmatory audit family (C2).** Now computed and reported (Section 5.8, Table 4): the four-member per-seed Holm family rejects the random-deferral null in all five seeds on MVTec AD — a constructive verdict, every bootstrap CI excluding zero. The single cross-seed rollup was authored post-freeze as a one-shot reduction but is moot, the per-seed confirmatory verdict being unanimous across seed-min, seed-median, and seed-max.

- **B3 (train-good quantile) practice.** Now completed post-hoc for PatchCore (Section 5.8): the confirmatory family treats B3's absence as the preregistered six-to-four degradation (PREREG §4), and, since a held-out train-good score pool exists for PatchCore, its B3 arm is reported post-hoc — constructive in all five seeds, consistent with B1/B2. B3-Dinomaly remains impossible (no train-side score dump), so the full six-member family cannot be run; on VisA no train-good scores were dumped for either backbone, so B3 is unavailable there.

- **V1 tier-2 as a verdict.** Now graded (Section 5.10, Table 5): under amendments A1/A2/A3 the MVTec escaped axis fails in 115/130 powered cells — the A2-expected estimator-variance behaviour, not a certificate failure, with tier-1 passing all 150 cells — and the false-reject axis is structurally ungraded (0/150; A1+A3). Its +3pp tolerance is near-unpassable for a correctly-calibrated gate by construction. VisA (exploratory) is partially gradeable on both axes.

- K6 (scoop re-scan). Now executed. The pre-submission citation re-scan for any prior certified three-way triage on MVTec AD or VisA was run fresh on 2026-07-13 and returned a clear verdict — no scoop (Section 2), so the pre-decided pivot to C2+C3-primary was not triggered. K6 is a *novelty* gate, not a statistical one: it gates on no frozen result, so it is correctly run last, immediately before submission. The 2026-07-11 freeze therefore predates this execution by design; running K6 after the freeze is the intended sequencing, not an inconsistency.

- K4 (audit headroom). K4 needs an oracle-headroom statistic the certification module does not yet implement.

- **Remaining tables/figures.** The confirmatory audit table (Table 4, now including the VisA post-freeze exploratory rows) and the tier-2 verdict table (Table 5, now a per-category breakdown) are now included. The defect-type Mondrian coverage-debt map (MVTec only; VisA has no defect-type axis) and the train-holdout G2 delta figure remain to be produced.

The latency/practicality slot flagged in the frozen skeleton is now computed and reported in Section 5.12 (Table 7): the gate adds a measured $2.8\,\mu$s/image on CPU on top of the backbone forward pass, confirming the O(sort) "negligible" claim with a real measurement rather than an argument.

Both reproduced backbones sit near ceiling on MVTec AD (mean image-AUROC 0.982 PatchCore, 0.996 Dinomaly), and Dinomaly is saturated enough per category that the exploratory excess-AURC audit has almost no headroom to detect practice skill on it there — its MVTec "0/30 reject" (with 14/30 degenerate cells) is uninformative about practice quality rather than evidence against it. The pre-freeze skeleton's designed answer was a lower-ceiling second benchmark (stated there as a plan; VisA is not part of the frozen preregistration), and it behaved as designed: on VisA (Section 5.7) the same audit has zero degenerate cells and rejects the random-deferral null in 14/24 (PatchCore) and 16/24 (Dinomaly) seed-0 practice-cells. The confirmatory pooled audit (Section 5.8) settles the deferral-skill question on MVTec AD in the constructive direction, and the post-hoc binding demonstration (Section 5.9) exhibits the still-sharper event directly: (backbone, category) cells where a naive fixed threshold's realized escaped-defect or false-reject rate violates α_{miss} or α_{fr} while the certified gate holds within target on the same data — the event the excess-AURC audit does not itself test. All three benchmarks are academic: no production-line, temporal-drift, or real-factory data backs the industrial framing here (threshold-transfer across production periods is future work), so the paper's claims are about certified triage on the standard inspection benchmarks, not about a deployed line.

Small good-calibration pools are intended behavior, not a shortfall. Under the primary protocol, G2's 4/15 certifiable count is a property of MVTec AD's test-good counts; honestly reporting it (Table 2) is the designed behavior, not a defect to be engineered away outside the frozen calibration-efficiency remedy (Section 5.4), and VisA's 12/12 under the identical arithmetic (with the identical code path) shows the refusals track the data, not a conservatism baked into the gate. MPDD makes the same point at the opposite extreme: its 0/6 primary-protocol G2 count is the honest, data-limited behaviour of uniformly small test-good pools, and the flag-gated train-holdout arm rescues it to 4/6 (one category KS-audited out; Section 5.5), so the three benchmarks together give three points consistent with pool-size-driven refusal ($0/6 \rightarrow 4/15 \rightarrow 12/12$). MPDD is post-freeze exploratory and its train-holdout rescue is PatchCore-only (Dinomaly dumps no train-side scores), disclosed as such and never entering the confirmatory family.

7. Conclusion

We described a certified three-way triage gate that turns any anomaly detector's scores into an auditable inspection decision with finite-sample escaped-defect and false-reject guarantees, and a refusal mechanism that makes "not enough data to certify" a visible outcome. On MVTec AD, VisA, and MPDD, across two backbones and five seeds each, the calibration machinery matches its preregistered arithmetic exactly, with no kill-gate tripped in any of the 30 (backbone, seed) aggregates. The three benchmarks exercise complementary failure modes and together give three points consistent with pool-size-driven refusal (0/6 → 4/15 → 12/12): MVTec AD stresses the refusal side (G2 certifiable in only 4/15 categories under the primary protocol, lifted to 12–13/15 by the frozen train-holdout remedy — PatchCore-only by data availability — with KS-failing categories honestly refused); VisA — lower-ceiling, larger good pools — certifies both axes in all 12 categories, cuts median deferral to 4–19%, and gives the excess-AURC audit the headroom MVTec's saturated backbones deny it; and MPDD (real painted-metal parts, post-freeze exploratory) is the stingy extreme, refusing G2 in all 6 categories under the primary protocol and rescued to 4/6 by the flag-gated train-holdout arm (one category KS-audited out). The confirmatory pooled audit (C2) returns a constructive verdict on MVTec AD — field-standard threshold practice earns real deferral skill over honest random deferral in all five backbone seeds (Section 5.8).

Data and code availability

MVTec AD [1], VisA [42], and MPDD [29] are publicly available benchmarks. Code and supporting materials for this study are archived at Zenodo (concept DOI: `10.5281/zenodo.21392290`; version 0.4.1) and maintained at `https://github.com/PeterPonyu/inspect-gate`.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. *[TODO-USER: confirm or amend this standard statement, and disclose any funding-related interests.]*

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CRediT authorship contribution statement

[TODO-USER: a CRediT statement matching the final author list, e.g. "Zeyu Fu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization."]

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